

Dp optimize(Quadrangle inequality)

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Description:

After introduce **optimization point** denoted as opt , we observe that in some situation the opt of dp transition is non-decreasing, we called that **quadrangle inequality**, and it can be show that if your cost function fullfill the Qi then the opt is non-decreasing, this lead us to redcut some complexity from $O(n^2)$ to $O(n)$ or $O(\log n)$

Quadrangle identity:

$$\begin{aligned}a &\leq b \leq c \leq d \\cost(b, c) &\leq cost(a, d) \\cost(a, c) + cost(b, d) &\leq cost(a, d) + cost(b, c)\end{aligned}$$

Algorithm:

Consider two kind of dp transition below:

$$dp[i][j] = \min(dp[k][j-1] + c(k+1, i)) \quad (1)$$

$$dp[i][j] = \min(dp[i][k] + dp[k+1][j] + c(i, j)) \quad (2)$$

(1) is one of the classic problem which is

Given n element and split them into k several continuous segment such that each segment don't overlap and their union form exactly original array

it can be show that if $c(x, y)$ fullfill the Qi then $opt(i, j) \geq opt(i-1, j)$, we can prove by contradiction and let k_1 and k_2 be opt of $dp[i-1][j]$ and $dp[i][j]$ and let $k_2 < k_1$, and since k_2 is opt of $dp[i][j]$ this imply

$$dp[i][j] = dp[k_2][j-1] + c(k_2+1, i) < dp[k_1][j-1] + c(k_1+1, i)$$

and we know that

$$\begin{aligned}k_2 &< k_1 < i-1 < i \\k_2+1 &< k_1+1 \leq i-1 < i\end{aligned}$$

so by Qi

$$\begin{aligned}
c(k_2 + 1, i - 1) + c(k_1 + 1, i) &\leq c(k_2 + 1, i) + c(k_1 + 1, i - 1) \\
dp[k_2][j - 1] + c(k_2 + 1, i) &< dp[k_1][j - 1] + c(k_1 + 1, i) \\
dp[k_2][j - 1] + c(k_2 + 1, i - 1) + c(k_1 + 1, i) - c(k_1 + 1, i - 1) &< dp[k_1][j - 1] + c(k_1 + 1, i) \\
dp[k_2][j - 1] + c(k_2 + 1, i - 1) &< dp[k_1][j - 1] + c(k_1 + 1, i - 1) \\
dp_{k_2}[i - 1][j] &< dp_{k_1}[i - 1][j]
\end{aligned}$$

the above result show that the opt for $dp[i - 1][j]$ is k_2 but it lead to a contraction, so $k_2 \geq k_1$

(2) is

Given n individual element and merge them until 1 element left, the cost value is the sum of the segment after merge

it can be show that if cost function fulfill the Qi, then

$$opt(i, j - 1) \leq opt(i, j) \leq opt(i + 1, j)$$

it can be show that if cost function fulfill the Qi then dp transition will also fulfill the Qi, the proof is too long so i didn't prove here, we just use the conclusion, let $opt(i, j - 1) = k_1$ and $opt(i, j) = k_2$ and $k_2 < k_1$ we will prove it by contradiction

$$\begin{aligned}
dp(k_2 + 1, j - 1) + dp(k_1 + 1, j) &\leq dp(k_1 + 1, j - 1) + dp(k_2 + 1, j) \\
dp(k_2 + 1, j - 1) + dp(k_1 + 1, j) &\leq dp(k_1 + 1, j - 1) + dp(k_2 + 1, j) \\
dp_{k_2}(i, j - 1) + dp_{k_1}(i, j) &\leq dp_{k_1}(i, j - 1) + dp_{k_2}(i, j) \\
dp_{k_1}(i, j) - dp_{k_2}(i, j) + dp_{k_2}(i, j - 1) &\leq dp_{k_1}(i, j - 1)
\end{aligned}$$

this imply k_2 is better option for $dp(i, j - 1)$ which is contradiction to our assumption thus $opt(i, j) \geq opt(i, j - 1)$, similar operation can apply on $opt(i + 1, j)$